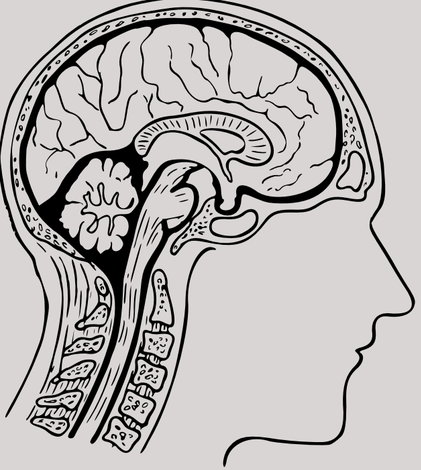
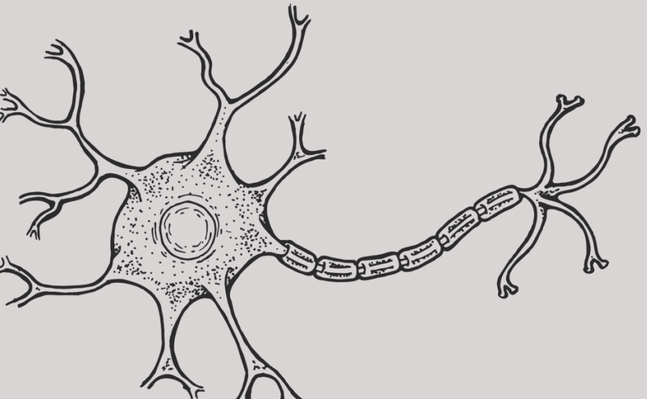
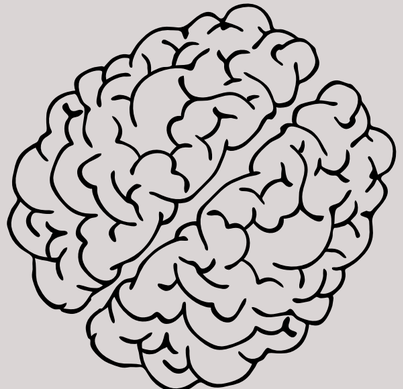


The Developing Human Connectome Project: A fast deep learning-based pipeline for neonatal cortical surface reconstruction

Qiang Ma, Kaili Liang, Liu Li, et al

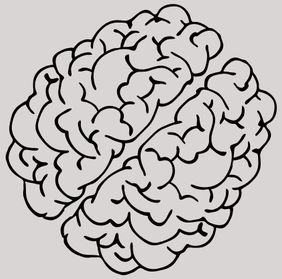
Medical Image Analysis





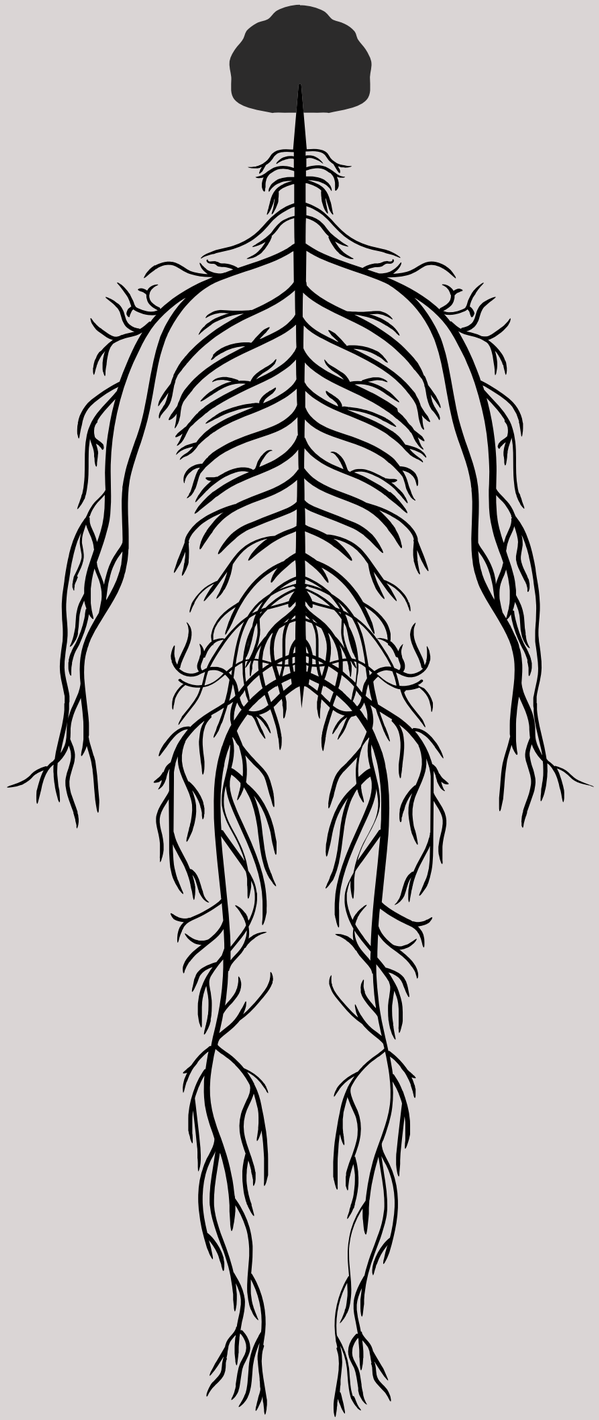
Why Neonatal Cortical Surface Reconstruction Matters?

- The brain develops rapidly during the perinatal period
- dHCP aims to study early brain development using MRI
- Cortical surfaces are important for analyzing brain structure and maturation
- Reliable neonatal cortical surface processing is therefore a key research need



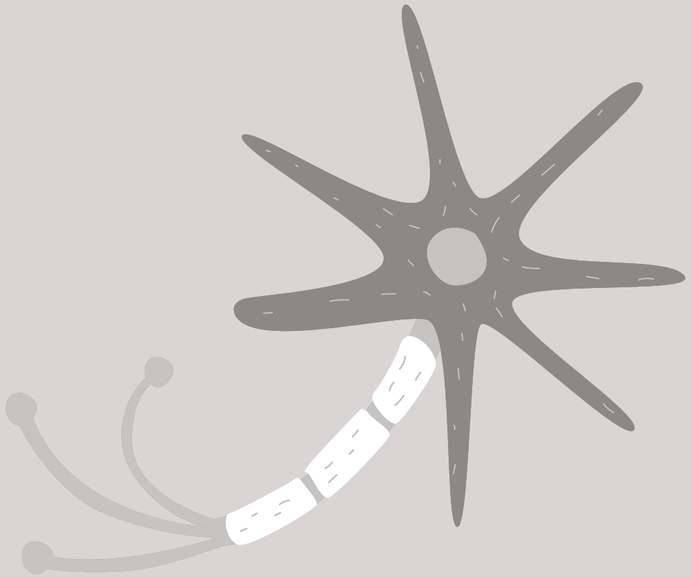
What Are Cortical Surfaces?

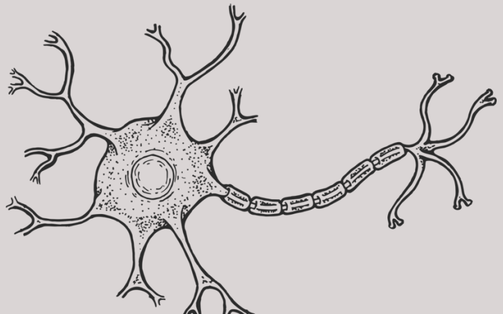
- White matter surface: boundary between WM and cortical gray matter
- Pial surface: boundary between cortical gray matter and CSF
- Represented as 3D surface meshes
- Should have sphere-like topology (genus 0)
- Used to measure:
    I. cortical thickness
    II. mean curvature
    III. sulcal depth



Why Neonatal MRI Is Challenging?

- Adult MRI pipelines do not transfer directly to newborn brains
- Neonatal MRI has limited resolution
- Neonatal brains are smaller
- Brain shape and size change rapidly with age
- MRI contrast is different from adults
- Therefore, T2-weighted MRI is commonly used





Existing Pipelines!

- Original dHCP structural pipeline
- Other infant pipelines
- DL pipelines based on implicit surfaces

i. Several hours per scan
ii. Strong dependence on segmentation quality
iii. topology correction is expensive
iv. Not ideal for large-scale studies

Main Idea

- Fast end-to-end neonatal surface pipeline
- Replace slow conventional steps
- Use DL + GPU acceleration

Key Contributions

- Multiscale diffeomorphic surface reconstruction
- No tissue segmentation for main surface reconstruction
- Faster spherical mapping
- 24 s total runtime

Dataset
Data Preparation

Dataset

783 subjects

887 scans

PMA 27–45 weeks

Templates

40-week atlas

Image template

Surface template

Sphere template

150k vertices

Training data

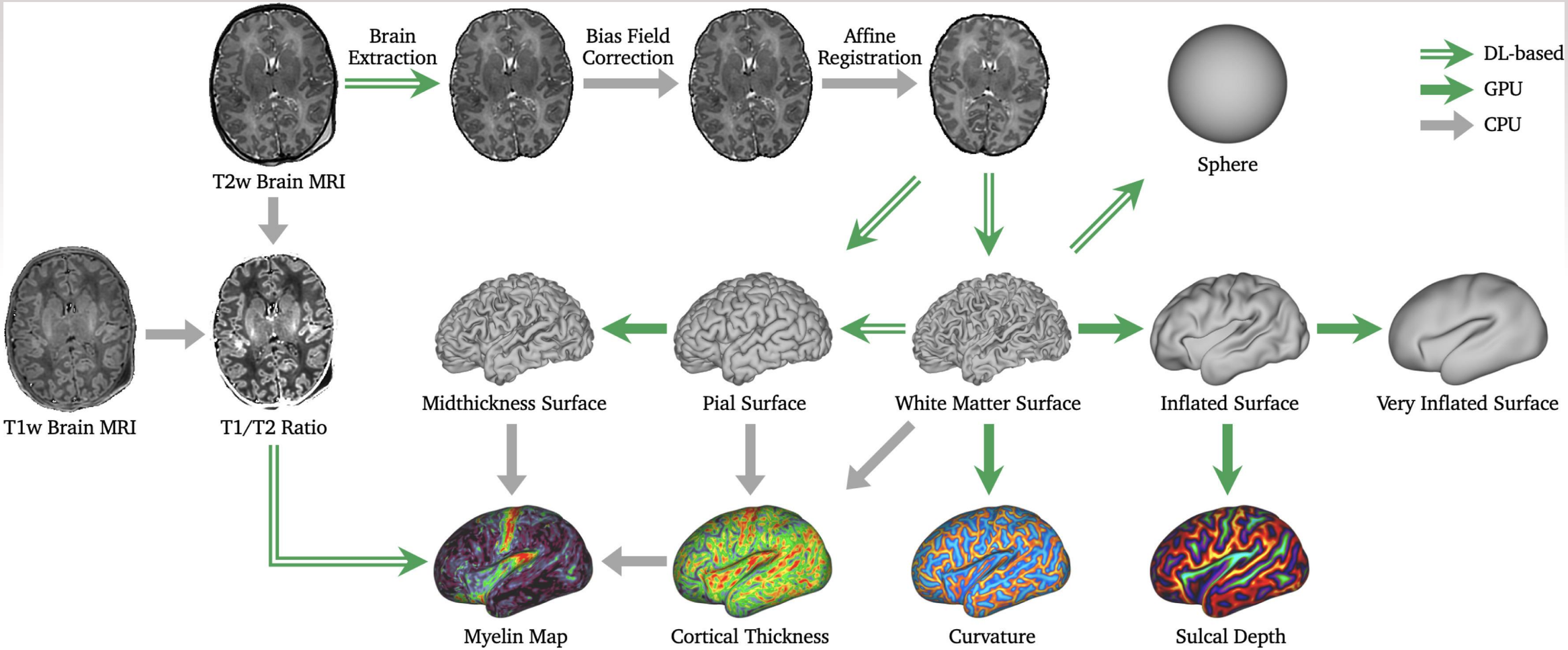
879 usable images

T2w MRI

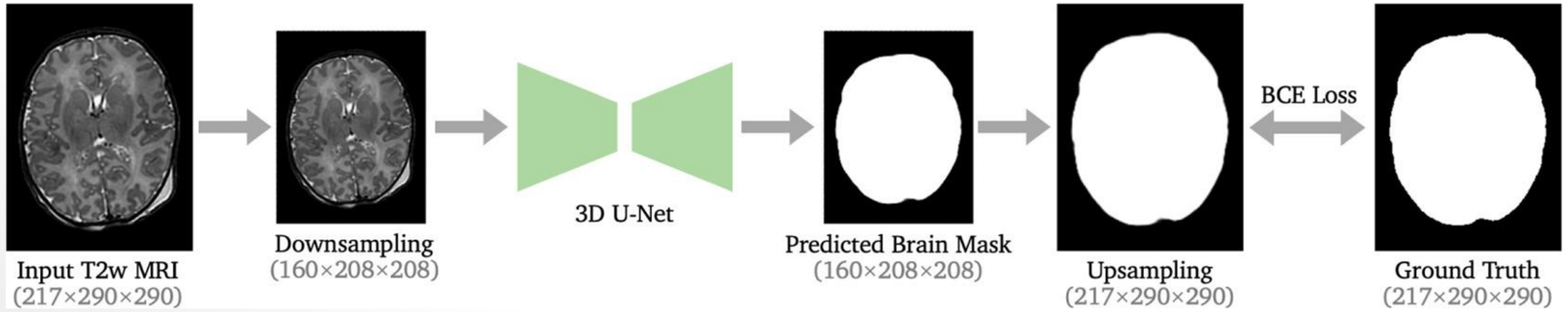
Pseudo-GT surfaces/segmentations

Overall Pipeline Overview

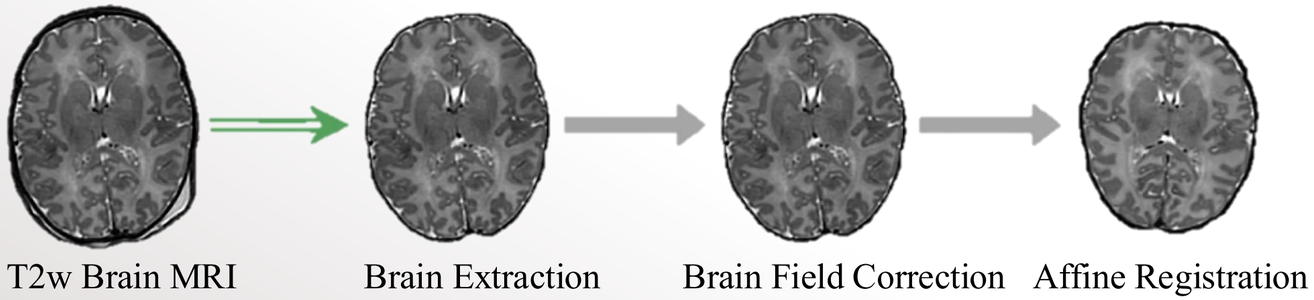
Preprocessing ❖Surface reconstruction ❖Surface inflation ❖Feature estimation ❖Spherical projection



i. Preprocessing



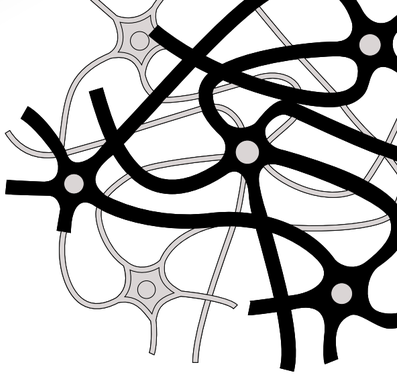
- Brain extraction
  - 3D U-Net predicts brain mask
  - Input downsampled to 160 × 208 × 208
  - Trained with BCE loss



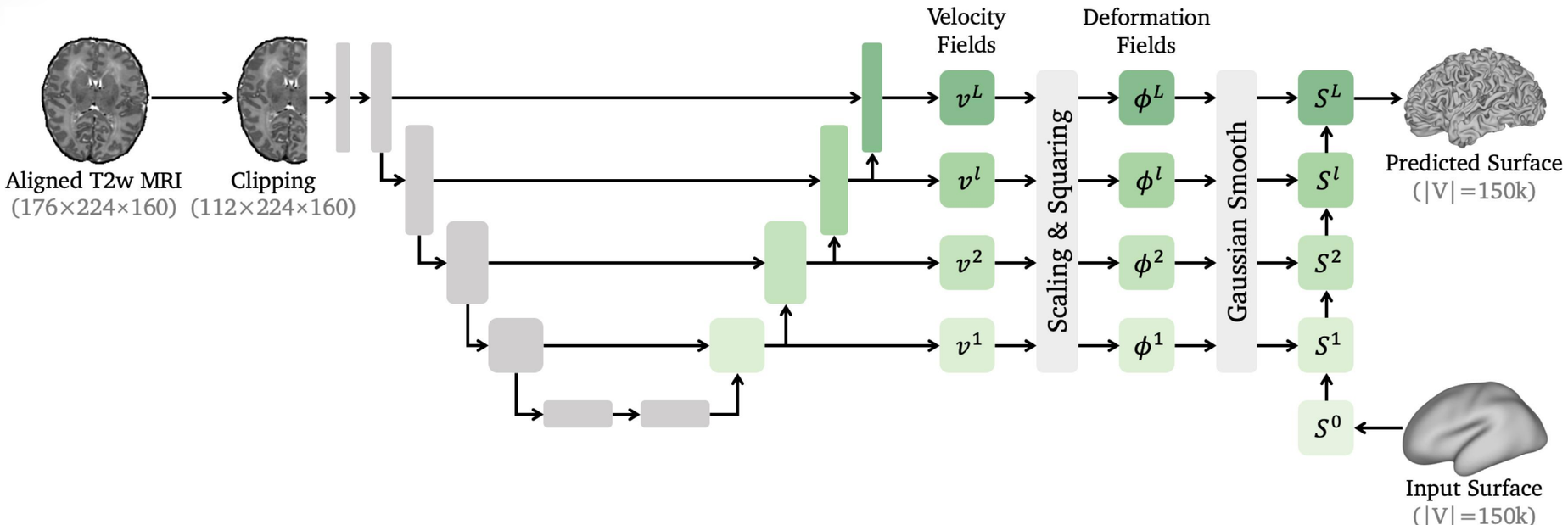
- Bias field correction
  - N4 normalization of T2w intensity

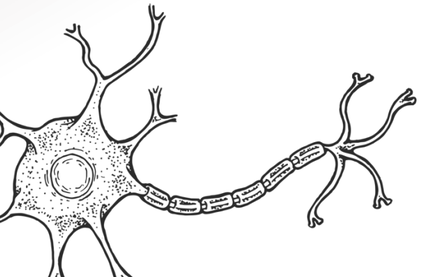
- Affine registration
  - Rigid + Affine alignment to 40-week dHCP atlas
  - Output template space: 176 × 224 × 160
  - Implemented with ANTsPy

ii. Cortical Surface Reconstruction



- End-to-end reconstruction from **aligned T2w MRI**
- Start from a shared **template surface**
- Learn **diffeomorphic deformations**
- Reconstruct **white matter** and then **pial** surfaces
- No tissue segmentation required as input





ii. Cortical Surface Reconstruction

Reconstruction network

Input: clipped hemisphere MRI (112 × 224 × 160)
3D U-Net predicts multiscale SVFs
SVFs integrated with scaling and squaring
4 scales: coarse → fine
ϕ_1, ϕ_2 :global shape
ϕ_3, ϕ_4 :cortical fold refinement

Training and losses

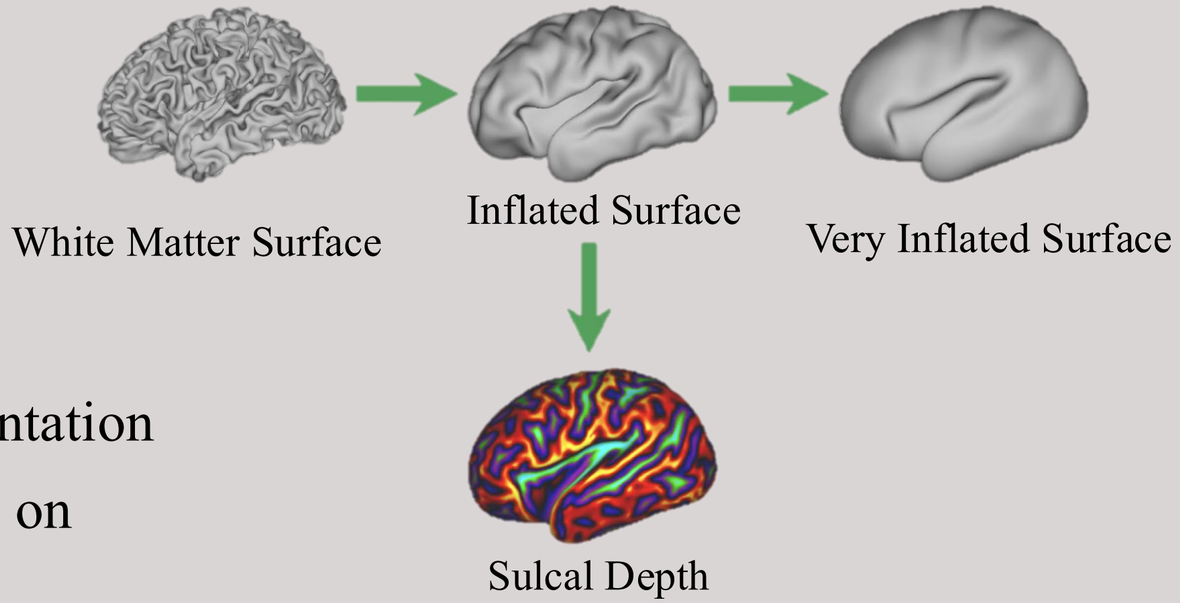
4 networks: LH/RH × WM/Pial

Loss:
- Chamfer distance → match target surface
- Edge loss → regular mesh edges
- Normal consistency → smooth surface
- Pseudo-GT surfaces from original dHCP pipeline
- Best model selected by validation Chamfer distance

iii. Cortical Surface Inflation

- Two inflation types are used

  Inflated / very inflated surfaces for visualization
  White matter surface inflation with minimized metric distortion



- Original tools replaced by PyTorch GPU implementation

  Sulcal depth is computed during inflation based on
  vertex displacement along the normal direction

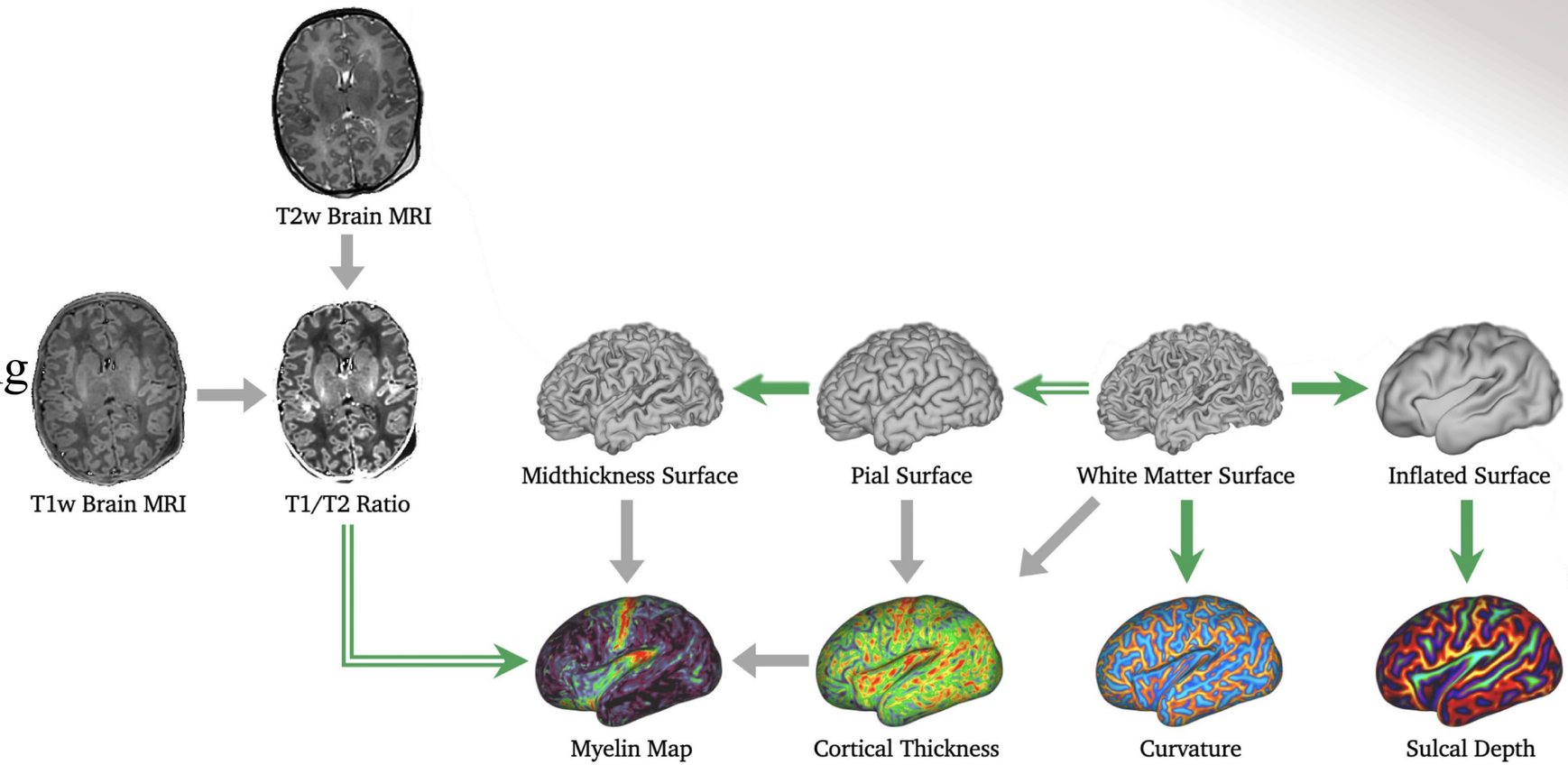
iv. Cortical Feature Estimation

Morphological features

Thickness
Curvature
Sulcal depth

Functional features

Myelin map
T1/T2 ratio
Volume-to-surface mapping
Cortical ribbon mask



v. Spherical Projection

Goal: map the white matter surface onto a sphere with minimal distortion

Use a **fixed initial sphere** shared across all subjects

Resample WM surface to an **icosphere** (**163,842 vertices**)

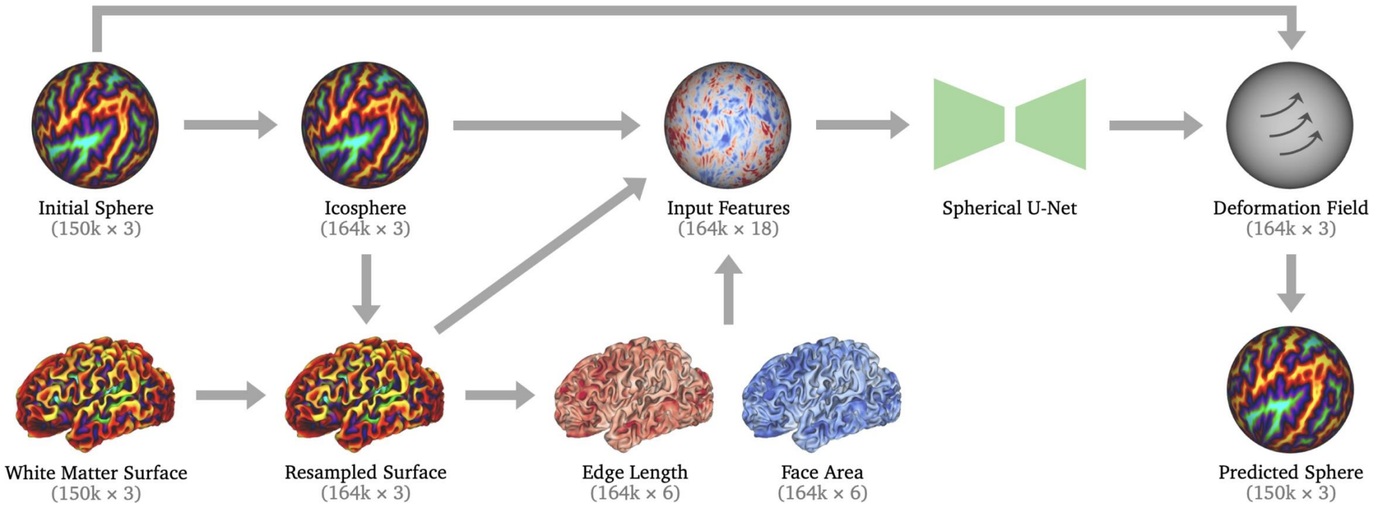
Input to Spherical U-Net:

surface coordinates
sphere coordinates
edge length
face area

Predict a **diffeomorphic** spherical **deformation**

Train with unsupervised edge + area distortion loss

Runtime: **0.33 s** per hemisphere



Implementation and Comparison to Original dHCP

Original dHCP: MIRTK + WB + FSL
Proposed pipeline: mainly Python + PyTorch

Key method changes

Brain extraction: FSL BET → U-Net
Tissue segmentation: Draw-EM → not required in main surface reconstruction
Surface reconstruction: deformable model → multiscale deformation network
Spherical projection: Spherical MDS → Spherical U-Net

Implementation details

PyTorch3D for mesh learning
ANTsPy for N4 + affine registration
WB for visualization and myelin map
GPU memory: 12 GB training, 7.6 GB inference

The number of self-intersecting faces (SIFs) on cortical surfaces.

Method	White matter surface	Pial surface
Original dHCP pipeline	0.457 ± 7.409	**0.517 ± 7.418**
Forward Euler	0.864 ± 5.712	4.925 ± 17.71
w/o Gaussian smoothing	0.613 ± 4.304	3.715 ± 9.824
Ours	**0.179 ± 3.171**	1.500 ± 8.875

Comparison to Other DL Pipelines

Compared with **FastSurfer** and **iBEAT** V2.0

Approx runtime:

FastSurfer: 1 h
iBEAT V2.0: 4 h
Ours: 24 s

Their pipeline is:

150× faster than FastSurfer
600× faster than iBEAT V2.0

Trade-off:

Does not yet provide tissue
Segmentation or cortical parcellation

Runtime:

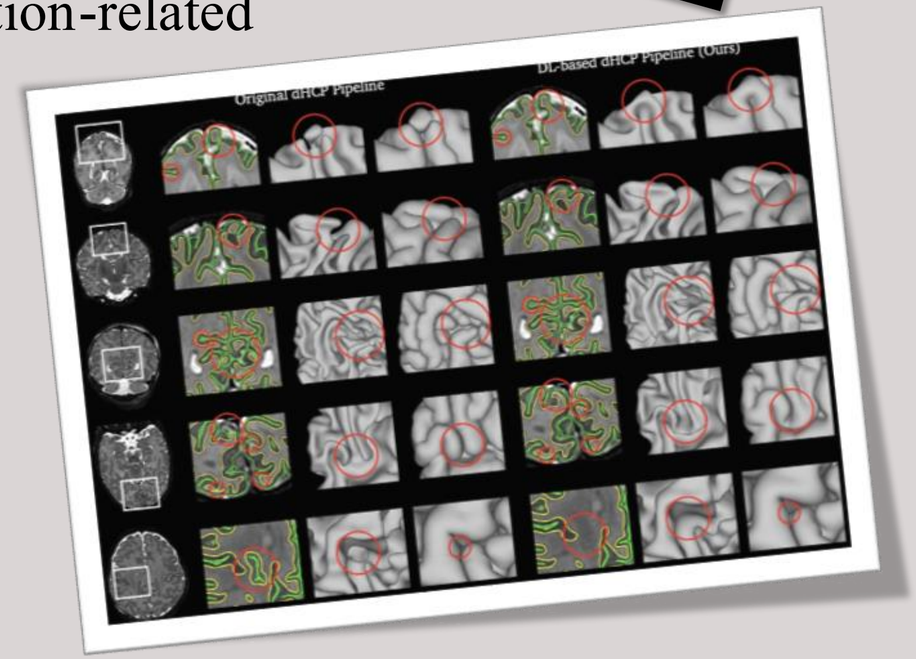
Original dHCP: 6 h 38 min
Ours: 23.98 s
995× faster

Results: Surface Quality and Generalization

Blinded manual QC on 30 test scans by 4 expert assessors

DL pipeline achieved superior or equal quality in 82.5% of cases
- 54.2% superior
- 28.3% equal

Main improvements in top / occipital regions and fewer segmentation-related artifacts such as holes/caves

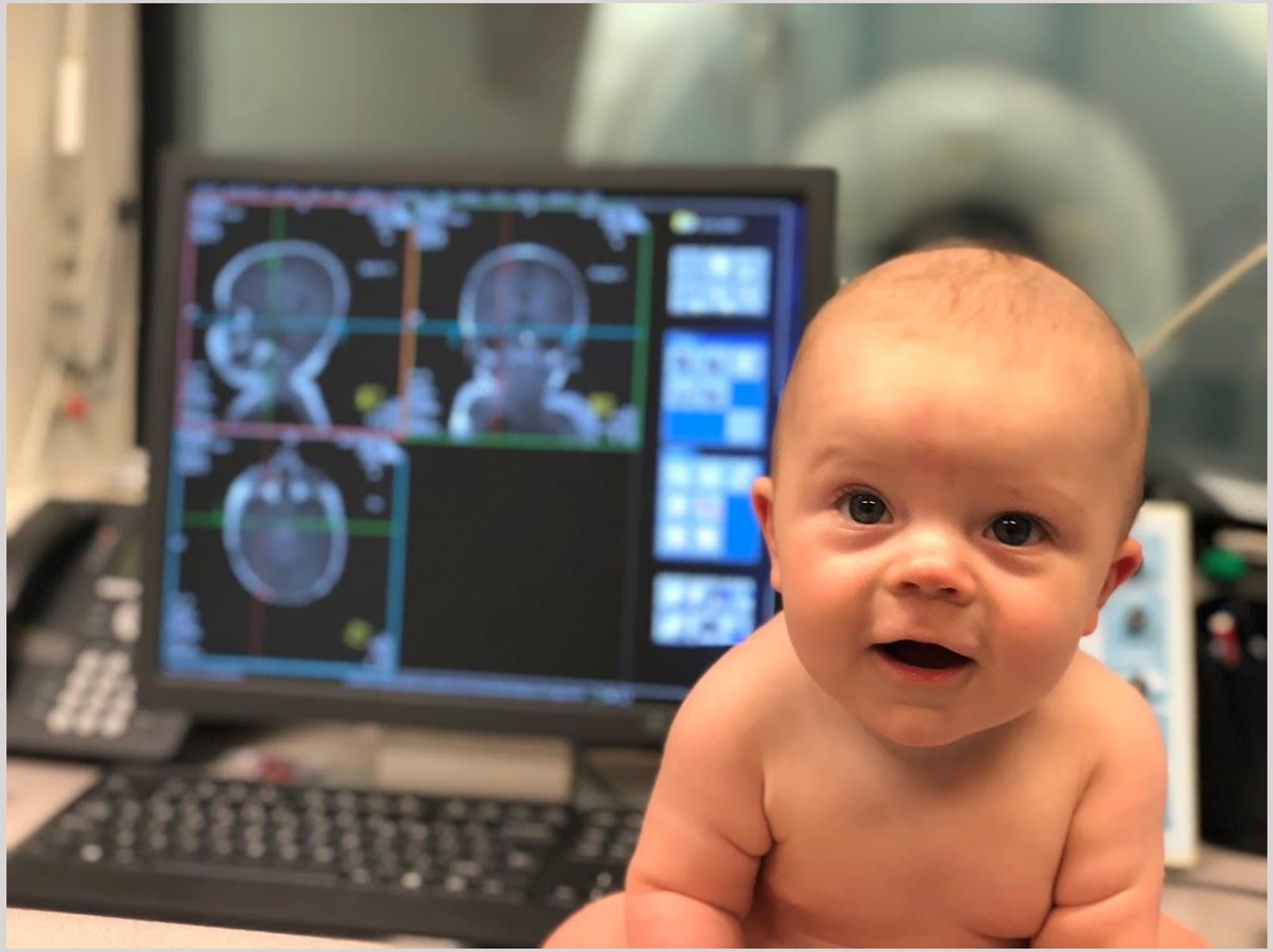


Captures expected brain-development trends with PMA
- stronger thickness correlation
- more age-consistent sulcal depth

Generalizes to unseen ePrime neonatal MRI without fine-tuning

Conclusion

- ✓ Fast DL pipeline for neonatal cortical surface reconstruction
- ✓ End-to-end T2w MRI → surfaces via multiscale diffeomorphic deformation
- ✓ Fast spherical projection + GPU-accelerated processing
- ✓ 23.98 s runtime, 995× faster
- ✓ 82.5% superior/equal visual quality
- ✓ Shows expected brain development patterns and generalizes to unseen data
- ✓ Limitation: no tissue segmentation/parcellation yet



Thank You!

Kaveh Moradkhani